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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Reconfigurable Radio Systems (RRS).

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Executive summary

The present document examines the feasibility and role of AI/ML techniques for Spectrum Sharing (SS). It analyses current regulatory signals and research trends from major regions (US, UK, EU, Australia, and selected Asia-Pacific markets), distils the state of the art in AI-enhanced spectrum awareness/sensing, dynamic allocation, and interference impact reduction, and identifies security and privacy risks together proposing possible mitigations appropriate for AI-enabled Spectrum Sharing.

The present document analyses policy trajectories across jurisdictions, and contemplates AI-assisted spectrum management (e.g. US National Spectrum Strategy and its 2024 Implementation Plan, Ofcom's *hybrid sharing* vision for upper 6 GHz), which make sharing viable in bands that are impractical to clear for exclusive assignment ([i.1] and [i.2]).

Technically, AI/ML improves detection accuracy, prediction, and real-time decision making, enabling improved protection of incumbents and a more efficient utilization of the shared spectrum. However, AI also widens the attack surface (data poisoning, spectrum spoofing, model extraction) and raises privacy concerns, underscoring the need for adversarial robust models, cryptographic protections, certification, and auditing.

Key conclusions:

i) AI/ML is a *material* enabler of Spectrum Sharing under diverse coexistence regimes;

ii) regulators increasingly plan for AI-augmented management;

iii) standardization should cover architectures, metrics, datasets, assurance, and conformance for AI-enabled SS; and

iv) security/privacy requirements are essential from the outset.

Introduction

The industry is at cross-roads in respect of spectrum needs and its availability. Continuous growth of capacity demand, especially in view of upcoming 6G deployments, highlights the need for additional spectrum. However, the more desirable part of available spectrum is fully saturated with very little possibility of exclusive usage. Thus, regulators are exploring new avenues for making the most efficient use of spectrum, including the possibilities of spectrum sharing.

The scarcity of desirable spectrum for 6G and the rising complexity of coexistence with non-3GPP incumbents (e.g. radars, satellites) make dynamic, automated sharing attractive. AI/ML offers concrete capabilities for enhancing sensing, prediction, optimization, and control that are impractical in fully algorithmic architectures. The US National Spectrum Strategy (NSS) and the 2024 Implementation Plan explicitly pursue technology-enabled sharing and band studies [i.3].

In Europe, Ofcom's 6 GHz *hybrid sharing* concept and BEREC's analyses of AI in telecoms point to future regulatory models where *sharing-by-design* is embedded in equipment standards. The EU **AI Act** will also shape deployment of AI in network functions, including spectrum functions [i.2].

The present document captures an early perspective of the possible impacts of AI on spectrum sharing in the foreseeable future.

Artificial Intelligence (AI) techniques are expected to impact the capabilities of transmitters and receivers of virtually all telecommunication services, and AI is also likely to have impact on similar communication services that use spectrum and the ways in which devices access and share spectrum.

AI thus can potentially manage transmissions in particular frequencies, in particular directions and at particular time-windows, thus reducing interference towards other users of the same spectrum. AI can also potentially allow reception of certain levels of interference by reducing its impact on the desired signal. Additionally, AI has the potential to improve interference estimates, spectrum occupation forecast and manages automatically and transparently dynamic spectrum access and sharing. All of the above may allow less conservative deployment scenarios to protect incumbents.

These enhanced capabilities may allow regulators to regulate and manage spectrum in ways that were unimaginable even a few years ago.

1 Scope

The present document studies technical approaches for automated spectrum access to support dynamic, temporary, and flexible spectrum sharing. In light of recent advancement on AI/ML and their potentiality, the aim of the present document is evaluating the possibility to include similar mechanism within a Spectrum Sharing framework. In particular, it analyses the possible advantages of such techniques, and their relative risks from both a technical and regulatory point of view.

The present document will begin by analysing the outlook for AI/ML based solutions for spectrum sharing among the different regions of the world.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] NTIA (March 2024): "National Spectrum Strategy Implementation Plan".

[i.2] Ofcom: "Consultation: Hybrid sharing: enabling both licensed mobile and Wi-Fi users to access the upper 6 GHz band".

[i.3] NTIA (November 2023): "National Spectrum Strategy".

[i.4] Autumn Statement 2023 - GOV.UK.

[i.5] Everything you need to know about Spectrum Sandboxes | UKTIN.

[i.6] "Deep Learning for Spectrum Sensing in Cognitive Radio Networks: A Survey", by S. Sharma, et al. (2020).

[i.7] "AI-Powered Spectrum Sensing for Cognitive Radio Networks: A Survey", by A. Al-Dweik, et al. (2021).

[i.8] "Predictive Spectrum Allocation for Cognitive Radio Networks using Machine Learning", by M. A. Khan, et al., IEEE™ Transactions on Cognitive Communications and Networking.

[i.9] "Deep Reinforcement Learning for Dynamic Spectrum Allocation in Cognitive Radio Networks", by A. Al-Shuwaili, et al., IEEE™ Transactions on Cognitive Communications and Networking, vol. 5, no. 1, pp. 123-135, 2019.

[i.10] "A Deep Reinforcement Learning Approach for Dynamic Spectrum Allocation in 5G Networks", by M. Chen, et al., IEEE™ Transactions on Wireless Communications, vol. 19, no. 10, pp. 6698-6712, 2020.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the following terms apply:

Artificial Intelligence (AI): field of computer science which enables systems/machines to simulate human intelligence in processes

Machine Learning (ML): subfield of AI which allows- systems/machines to imitate intelligent human behaviour through learning (identification of patterns)

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

AI	Artificial Intelligence
AoA	Angle of Arrival
API	Application Program interface
CBRS	Citizen Broadband Radio Service
CSMA	Carrier Sense Multiple Access
DNN	Deep Neural Network
DoD	Department of Defence
DP	Differential Privacy
DSIT	Department for Science, Innovation and Technology
ESC	Environment Sensing Capability
EU	European Union
FCC	Federal Communications Commission
GPT	Generative Pre-trained Transformers
IMT	International Mobile Telecommunication
LBT	Listen Before Talk
LSTM	Long Short-Term Memory
MIMO	Multiple Input Multiple Output
ML	Machine Learning
NOI	Notice Of Inquiry
NSS	National Spectrum Strategy
PCA	Principal Component Analysis
PU	Primary User
QoS	Quality of Service
RL	Reinforcement Learning
RNN	Recurrent Neural Network
RSSI	Received Signal Strength Indicator
SAS	Spectrum Access System
SINR	Signal to Interference plu Noise Ratio
SS	Spectrum Sharing
SU	Secondary User
SVDD	Support Vector Data Descriptor
SVM	Support Vector Machine
TAC	Technological Advisory Council
WRC	World Radiocommucation Conference

4 Survey of key administrations

4.1 Introduction

Spectrum regulations historically depended on a steady supply of the valuable resource and spectrum availability for users has so far been mostly stable during the period the regulations have been in force. However, as demand for capacity grows and desirable spectrum remains a scarce resource, every effort is underway to make the most efficient use of it. In this context, Artificial Intelligence (AI) and Machine Learning (ML) have become very important. Below are the basic definitions for some context:

Artificial Intelligence: The field of computer science which enables systems/machines to simulate human intelligence in processes.

Machine Learning: The subfield of AI which allows systems/machines to imitate intelligent human behaviour through learning (identification of patterns).

AI/ML techniques are now used widely in radio technologies including in AI-native air interfaces, for various purposes including channel estimates, scheduling and resource allocation, interference cancellations, load balancing/traffic steering, etc.

Two recent AI/ML technology trends may significantly impact the regime of spectrum management and regulation. First, there has been a revival of traditional AI over the past decade, driven by an explosion of availability of different types of data, which in turn, was enabled by tremendous growth in storage and computational technologies. This AI boom has been further brought to the forefront of technology discussions by the maturity of generative AI in the last couple of years as reflected in advancements in Generative Pre-trained Transformers (GPT) in applications such as OpenAI's ChatGPT® or Google's® Bard/Gemini. These fundamental models are being extended for numerous customized models to suit numerous purposes in virtually any application in any industry.

Secondly, with 6G in the horizon, the search for new spectrum to adopt the new mobile generation and to accommodate the continued explosion of mobile data has started all over again. However, the required amount of exclusively licensed spectrum in desired frequencies is virtually non-existent as the entire range is occupied by critical applications in most developed countries. Hence there is a renewed interest in spectrum sharing - especially vertically between mobile operators and non-3GPP-based incumbents, e.g. radars, microwave radios, satellites etc. This notion of spectrum sharing is opening new horizons for using AI, which in turn, provides administrations new tools for making spectrum available.

Regulators in the US, UK and Europe are envisioning that AI will be a key technology to realize the possibilities of efficient spectrum management and sharing in the 6G era. The only wide-scale implementation of spectrum sharing in the world, CBRS band in the US, has been a rather challenging exercise to gain eventual commercial success. Newer AI/ML technologies promise significant improvements over the current state of the art.

Analysis Mason in a recent publication discussed how "AI-techniques" may be used for spectrum management but concluded that "given the complex nature of interference avoidance and the limited amount of experience in using AI/ML techniques, machine learning seems unlikely to result in significant overhauling of general spectrum management approaches. However, it is possible that dynamic forms of sharing within specific bands designated for such purposes could be optimized through devices and networks that use machine learning".

While there is a very broad impact of AI on mobile networks (e.g. traffic management, performance enhancement, Quality of Service (QoS) delivery, etc.) and their operations (e.g. energy consumption, maintenance, customer service, etc.), the discussions in the present document are **restricted only to** the aspects that may impact **spectrum administration, management and regulations** in the foreseeable future. It is conceivable that newer AI applications will emerge that will impact spectrum usage and regulation - either directly or indirectly - and the present document will need to be updated at appropriate time intervals.

4.2 United States

The National Telecommunication and Information Administration (NTIA) of the United States published its National Spectrum Strategy (NSS) in November 2023, highlighting the need for using advanced technology techniques like **AI/ML** to maximize the utilization of the **valuable spectrum** resources**,** and the importance of **spectrum sharing**, given that desirable spectrum bands for 6G have already been allocated and in use, and there is not much room left for spectrum clearance to allow for exclusive 6G usage.

Earlier in the year the Federal Communications Commission (FCC) issued a Notice Of Inquiry (NOI) to assess the spectrum usage by various non-federal entities. This involves collection of a vast amounts of data and processing them, which would have been practically impossible even a few years ago, especially without today's advancements of AI/ML. This has been stated as:

> *"As the radiofrequency environment becomes more congested, leveraging technologies such as artificial intelligence to understand spectrum usage and draw insights from large and complex datasets can help facilitate more efficient spectrum use, including new spectrum sharing techniques and approaches to enable co-existence among users and services."*

In July 2023, the FCC and National Science Foundation (NSF) held a conference "AI for Communications Networks and Consumers". Chairwoman Rosenworcel mentioned that *'AI can be used to improve spectrum efficiency in the increasingly congested radio frequency environment by teaching our wireless devices to manage transmissions on their own, thus enabling the best use of spectrum without the use of a central, spectrum-sharing administrator.'* Commissioner Starks noted that *"he is considering how AI can be used to improve US wireless networks. In particular, he is considering how AI can be used to predict which frequencies will be used, where they will be used, and how long they will be used as spectrum availability becomes scarcer. He is also considering how AI can enhance wireless network planning by making virtualized deployments more efficient."*

In February of 2024, a new Technological Advisory Council (TAC) was announced for the FCC, and it has two separate workgroups assigned to AI/ML and Advanced Spectrum Sharing. They are expected to address using AI for efficient use of spectrum and their studies are likely to influence future spectrum management policies.

The above developments in the past few months clearly show renewed interest from FCC and NTIA in pursuing spectrum sharing in the new era of AI/ML. The National Spectrum Strategy Implementation Plan released in March 2024 outlines various aspects of spectrum management where advanced technologies involving AI/ML is likely to be used.

In 2025, however, the policy pivot in 2025 replaces a default preference for *dynamic, real-time sharing* with a hybrid strategy that prioritizes structured reallocation and exclusive licensing for large contiguous mid-band blocks while retaining sharing frameworks (e.g. SAS/ESC) where incumbency or mission constraints make full reallocation impractical. Concretely, the legislation restoring FCC auction authority and establishing a formal spectrum pipeline mandates the identification and repurposing of hundreds of megahertz (targeting at least 800 MHz of commercial spectrum, with ~500 MHz expected from federal holdings) and directs NTIA/FCC to sequence studies, transition plans, and auctions to minimize operational risk to DoD and other incumbents. Technically, this means regulators will favour exclusive-use, interference-limited block assignments (with guard bands, relocation funding, and transition timelines) over expanding aggregate-interference thresholds in incumbent-protected sharing regimes; where sharing remains, SAS/ESC systems will be treated as interim coexistence mechanisms during transition phases rather than the long-term default. The pipeline approach also imposes auction design constraints and revenue targets that shape band selection, reserve rules, and geographic licensing granularity, forcing technical trade-offs between spectral contiguity (for high-order MIMO and carrier aggregation) and incumbent protection (sensor density, exclusion zones, and conservative propagation models).

4.3 United Kingdom

In July 2023, Ofcom released a consultation on "hybrid sharing" (also known as Shared Use) of spectrum between licensed mobile and Wi-Fi users in the upper 6 GHz band. For the sharing scenarios considered, Ofcom wanted to hear the industry's views on potential solutions, including ideas for a possible combination of database and spectrum sensing mechanisms, for which the use of AI can significantly improve performance.

In November, the UK government held a major AI Safety summit. While the event was centred around AI for spectrum management, it highlighted the acknowledgement by the top governments and technology experts around the world that AI is going to have major impact on every aspect of our lives. Consequently, the UK's Department of Science, Innovation and Technology, through funding announced in the government's Autumn Statement [i.4], issued an invitation for tender under DSIT's spectrum sandboxes program, funding innovative spectrum sharing solutions which would help the government to shape new spectrum regulatory approaches [i.5]. Thus, in the broader sense, the UK government's efforts in using AI/ML in different aspects, including spectrum regulation, is evident.

4.4 European Union

The CEPT brief on WRC-23 Agenda Item 1.2 includes references to many sharing scenarios: *"the IMT Resolution clearly outlines opportunities for other broadband applications in the mobile services (i.e. WAS/RLAN) as well as sufficient flexibility regarding the future wireless broadband usage, i.e. by IMT, WAS/RLAN or under a shared framework between IMT and WAS/RLAN…"*.

BEREC, i.e. the Body of European Regulators for Electronic Communications, published a Report on the impact of AI solutions in the telecoms sector on regulation seeking to identify AI developments in telecom use cases to assess their potential impact on privacy, security, monitoring and eventually telecom regulation.

The European Union passed a landmark AI regulation law to manage the adoption of AI and control its risk of abuse. It is expected that AI-related technology will be used widely in the Union, including in matters of spectrum sharing regulation.

While some of these developments are not directly related to regulations pertaining to the telecom industry, they are expected to have indirect impact and thus captured in the present document. The AI regulations will likely encourage the industry to adopt the technology more widely and the administrations to develop regulations that will take into account the possible technological advancements.

A few examples are provided for illustrative purposes. Some countries have considered spectrum sharing in the 26 GHz band. Most recently, the Upper 6 GHz band sharing has also entered into discussions for sharing/coexistence purposes. It is quite conceivable that AI will be used to develop solutions that would not have been so effective without the underlying technology.

4.5 Australia

Australia's defence department is also developing advanced spectrum management systems. *"Distributed Autonomous Spectrum Management (DUST) uses machine learning and artificial intelligence to dynamically plan and allocate spectrum usage to achieve optimized spectrum utilization in congested, contested and complex environments"*. It is further clarified: *"The technology will be incorporated into capability projects for the Australian Defence Force"*.

5 AI/ML for spectrum sharing

5.0 Introduction

This clause introduces the main functional block that can enhanced using AI/ML techniques. In particular, spectrum awareness, sensing, allocation and interference reductions, are examples of systematic blocks that can be enhanced using AI/ML techniques. Together, such blocks can allow a more efficient sharing of the available spectrum.

5.1 Spectrum awareness and spectrum sensing

5.1.1 Spectrum awareness

Spectrum awareness relates to the capabilities of intelligent systems to learn and understand the spectrum environment. AI/ML can enhance this function by encompassing a range of capabilities, spanning from real-time spectrum monitoring to dynamic channel allocation using, for example, interference awareness techniques. The goal of this approach is to gain insights from vast datasets, yielding a deeper understanding of spectrum usage patterns, identifying potential bottlenecks, and predicting future demand. More in particular, these systems can detect the presence of various signals, identify their sources, and analyse their characteristics in such a way that, by continuously learning from the data, AI/ML can help predicting spectrum usage patterns, and make real-time decisions to optimize spectrum allocation.

More in detail, spectrum awareness refers to devices' capacity to *understand* the spectrum environment. In particular, this includes:

- **Available spectrum:** Which frequency bands are currently unused or underutilized.
- **Spectrum usage patterns:** How spectrum is being used over time (e.g. peak hours, specific locations).
- **Spectrum regulations:** Understanding the rules and regulations governing spectrum use in a particular region (e.g. maximum peak power).

The purpose of spectrum awareness is to make more efficient use of the available spectrum, which involves:

- **Dynamic spectrum access:** Devices can dynamically adjust their transmission parameters (frequency, power) to avoid interference with other users.
- **Cognitive radio:** Devices can "learn" about the spectrum environment and adapt their behaviour to optimize performance.

Spectrum awareness typically relies on:

- **Passive monitoring:** Devices *listen* to the spectrum to detect signals without transmitting.
- **Database access:** Accessing information about spectrum usage from centralized databases.

Currently, functions of spectrum awareness are not present in commercial devices but only envisaged at research level.

5.1.2 Spectrum sensing

Spectrum sensing concerns the process of detecting the presence of radio signals in the electromagnetic spectrum, and it is a is a fundamental component of intelligent spectrum sharing. Its role is simply to detect the presence or absence of primary users (e.g. licensed networks) in a given frequency band, with the goal of avoiding generating interference. An example of spectrum sensing devices can be found in CBRS, where the ESCs notify the presence of Navy radar.

Traditional spectrum sensing methods often rely on simple energy detection or matched filtering techniques, which can be susceptible to noise and interference. As a consequence, such systems are prone to false alarm or missed detection. However, the advent of AI and ML has opened new possibilities for more sophisticated and robust spectrum sensing [i.6].

In particular, machine learning algorithms can be trained on large datasets of spectrum data to learn the signature of different signals and therefore identify patterns that traditional methods are unable to detect. For example, deep learning models can be used to classify different types of signals based on their time-frequency characteristics, enabling more precise spectrum awareness [i.7].

Furthermore, AI/ML based spectrum sensing have the ability to adapt to dynamic spectrum conditions, learning to identify and track the presence of new signals and changing interference patterns yielding a more efficient spectrum sharing with devices that are able to dynamically adjust their transmission parameters to avoid interference and maximize their use of available spectrum.

Different techniques can be used:

- **Supervised Learning:** it involves training models on labelled data to identify patterns and predict spectrum occupancy. For example, Support Vector Machines (SVM) and neural networks can be adopted to train devices to recognize specific waveform.
- **Unsupervised Learning:** used for clustering and anomaly detection in spectrum data without labelled examples. Techniques like K-means clustering and Principal Component Analysis (PCA) are employed. Its main advantage is to be able to adapt to unknown primary users' waveforms.
- **Reinforcement Learning (RL):** algorithms learn optimal sensing strategies through trial and error. Techniques such as Q-Learning are used to develop adaptive sensing policies.

Globally, the adoption of AI/ML spectrum sensing can drastically reduce false positive/negative, by adapting the sensing the primary users' waveform. Dynamic threshold adjustment can be used to adjust the threshold for signal detection based on the spectrum condition, yielding a more precise and automated sharing policy.

5.2 Dynamic spectrum allocation

Dynamic spectrum allocation refers to capabilities of a shared spectrum system to quickly allocate and release spectrum from non-primary users. Currently, systems like CBRS SAS utilize a central controller and a fixed set of rules and parameters to allocate spectrum for non-primary users and demand the release of the spectrum following the appearance of a primary user. AI/ML can enhance these functions by increasing the flexibility of the central controller by substituting fixed rules and parameters with intelligent learning.

Dynamic Spectrum Allocation: the AI/ML advantage

AI/ML capability to analyse vast amounts of data and therefore identifying patterns, can be exploited in this context to enable the following aspects, enabling:

- **Predictive Modeling:** AI/ML can utilize historical data in order to anticipate future spectrum demands and availability, allowing for proactive spectrum allocation and ensuring that resources are available [i.8].
- **Real-time Optimization:** AI/ML algorithms can continuously monitor spectrum usage, adjusting allocations in real-time based on changing conditions. This ensures optimal resource utilization, minimizing wasted spectrum and maximizing network performance [i.9].
- **Adaptive Spectrum Sharing:** AI/ML can dynamically adjust spectrum access rights based on user needs, traffic patterns, and interference levels. This allows for flexible sharing between different user tiers ensuring fair and efficient access [i.10].

5.3 Interference impact reduction

Interference increase is the main limit to spectrum sharing. In shared spectrum systems, interference generated by non-primary users toward primary users should be kept below a strict threshold in order to not undermine primary users' functionalities. On the other hand, non-primary users to non-primary users and primary users to non-primary users' interference can severely reduce spectral efficiency. Interference cancellation refers to techniques mitigating the effects of unwanted signals that can disrupt or disturb the transmission and reception of information. There exist a number of traditional techniques relying on rules-based algorithms. For example, MIMO receiver can apply spatial filtering in order to avoid interference generated by specific direction, and protocols such as CSMA and LBT are used to avoid collision in an interfered channel. Techniques such as successive interference cancellation offer a more non-linear approach to interference reduction but require knowledge about the interferer transmission. Conversely, AI/ML offers a paradigm shift, enabling systems to learn and adapt to interference.

The role of AI/ML in interference cancellation

AI/ML techniques have the ability to impact all aspects of interference management: interference creation reduction, interference reception reduction, interference impact management:

1) Interference creation reduction: Traditionally, interference reduction at the transmitter side is achieved by correctly setting transmission power, beamforming, and avoiding specific transmission bands and time slot. The adoption of AI/ML allows to enhance, and expand, such techniques as follows:

    a) Collaborative spectrum sharing: AI/ML can enable collaborative spectrum sharing among multiple users by coordinating their activities (e.g. directions of transmission and reception) and minimizing mutual interference. This is particularly useful in dense network environments.

    b) AI/ML based beamforming: AI/ML can be used to modify beams or precoding vectors and matrices in order to reduce interference. This can be done by either shaping and modifying the single entries of the amplification and phase, or by increasing and decreasing the probability of each beam. One seeming implementation could be exploiting Reinforcement Learning (RL): beams or precoding matrices can be modified based on information on the generated interference. Interference can be considered as the only reward.

    c) AI/ML based power control: AI/ML can be used to increase and decrease transmission power, based on position and other characteristics of the transmitter. A simple reinforce based, e.g. on the received interference level can help an RL agent to tune the optimal power.

    d) Real-time optimization: RL agents continuously adapt to changing network dynamics, making real-time adjustments to minimize interference while maximizing spectrum utilization.

2) Interference reception reduction:

    a) AI/ML spatial adaptive filtering: AI/ML can be exploited to learn interference source patterns (e.g. position of the transmitters, AoA of the disturbing signal) and can select the correct receive beamforming matrix (i.e. spatial filter) to maximize the received SINR or minimize the received interference. AI/ML can learn the dynamic characteristics of interference and adjust filter parameters in real-time. Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Reinforcement Learning (RL) show promise for adaptive filtering.

3) Interference impact reduction:

    a) Deep learning interference cancellation: This technique adopts DNNs to isolate desired signals from complex noise environments (e.g. speech from background noise). The DNN can be trained on massive datasets of noisy and clean signal to learn complex representations.

    b) Blind source separation with AI: The basic idea at the basis of this technique is to separate mixed signals without prior knowledge of the original sources. Independent Component Analysis (ICA) through ML uses statistical techniques and ML algorithms to identify and separate statistically independent sources.

6 Security aspects of AI/ML spectrum sharing

6.1 Introduction

In most of the Spectrum Sharing scenarios, the security requirements include confidentiality, availability, integrity, authentication, compliance, access control, and privacy:

- Confidentiality: sensitive information should be disclosed solely to authorized entities and users, in particular in database-based Spectrum Sharing systems.
- Availability: devices should be able to access the spectrum when required and there is physical availability.
- Integrity: information exchanges should be protected from tampering and modification, and from malicious interactions such as alteration, insertion or cancellation.
- Authentication: devices should be allowed to establish secure connections and to verify their identity.

- Compliance: user behaviour should be compliant with various regulations and decisions of the Spectrum Sharing central entity in order to avoid disturbance such as harmful interference.
- Access control: only users and devices with credentials should be able to access the shared spectrum.
- Privacy: private information should be protected and should be shared only when necessary and allowed.

Possible risks to AI/ML enhanced spectrum shared mechanisms can arise from both the AI/ML aspect and the spectrum sharing system. In clause 6 possible attacks concerning the AI/ML decision making engine of AI/ML spectrum sharing mechanism are listed with their possible countermeasures.

6.2 Adversarial attacks on ML models

6.2.1 Data poisoning

Data poisoning refers to an attack technique in which an adversary corrupts the training or online learning data, if available. For example, sensing reports, rewards, gradient updates, could be maliciously transmitted to central Spectrum Sharing decision making entity in order to force the model into making persistently wrong decisions, such as considering a portion of the spectrum idle or busy, or force handoff decisions. This stealthy attack can slowly degrade performance, cause unintended PU interference, or even trigger a denial-of-service for SUs:

- **Risk:** Malicious users inject into the system maliciously crafted training samples, or feedback false information (e.g. falsified spectrum records) so that a classifier learns incorrect information (e.g. occupancy) and separation boundaries.
- **Impact:** Compromised spectrum release decisions, wrong channel availability estimation, widespread interference or denial of service.

6.2.2 Spectrum poisoning

Spectrum poisoning refers to a scenario in which a malicious user manipulates the input features, such as RSSI, modulation patterns, or cyclostationary signatures, fed into the Spectrum Sharing central controller's spectrum sensing model. The goal is to trick the classifier into misidentifying the spectrum state, for example, labelling a busy channel as idle or vice versa:

- **Risk:** At runtime, malicious users transmit over-the-air perturbations that mimic the PU footprints. They can cause the sensing model to misclassify idle channels as busy (or vice versa).
- **Impact:** Forced handoffs, throughput collapse, wasted sensing/transmission cycles.

6.2.3 Model extraction and inference

In a model extraction attack, a malicious user can interact with the spectrum sharing central controller, typically via API queries, with the goal of reconstructs its internal AI model. The goal is to predict its behaviour under various spectrum conditions. By analysing further the inferred model, the attacker can deduce sensitive information about the users such as PU and other users' location:

- **Risk:** Malicious users query the shared API (e.g. a spectrum access predictor) and, via black-box probing, reconstruct model parameters or training data characteristics.
- **Impact:** Leakage of PU activity patterns, SU behavioural profiles, enabling targeted jamming or inference of private user locations.

6.3 Privacy risks in AI/ML enhanced spectrum sharing systems

6.3.0 General

Given the aforementioned type of attacks, the following possible risks in an AI/ML enhanced or based spectrum sharing system can be listed.

6.3.1 PU location and usage inference

Through model extraction, and therefore frequent database queries by SUs, even if anonymized, can be correlated via differential-time fingerprinting to pinpoint incumbent radar or military installations.

6.3.2 SU identity and mobility tracing

AI/ML models trained on aggregated sensing logs may leak individual SU's location trajectories through model inversion.

6.4 Mitigation strategies

6.4.1 Adversarial training

Adversarial training is a defence technique aiming at strengthening AI/ML models against evasion and data poisoning attacks. This is done by deliberately training the model with perturbed or malicious inputs, that is feeding the model synthetic or worst-case examples, such as emulated primary user signals or spoofed sensing data. The goal is for the AI/ML model to learn to maintain correct classifications even under adversarial conditions.

6.4.2 Certified defences

Certified defences are a class of techniques aiming at improving the AI ML based spectrum sharing central controller resistance against small, adversarial perturbations, such as those used in evasion or data poisoning attacks. In the context of spectrum sharing, certified defences ensure that a classifier's decision, e.g. whether a channel is idle or busy, remains unchanged even if the input features, like RSSI, modulation patterns, or cyclostationary signatures, are slightly manipulated. One possible used method to obtain this is randomized smoothing, which wraps the original model with a probabilistic layer that averages predictions over noisy versions of the input.

6.4.3 Outlier and anomaly detection

Outlier and anomaly detection is a defence strategy aiming at identifying and filtering malicious or statistically suspicious inputs (data poisoning). Those inputs include among others possible falsified sensing reports, spoofed RSSI values, or manipulated gradient updates. The overall idea is to classify legitimate inputs by exploiting mechanisms such as Support Vector Data Description (SVDD), autoencoders, or clustering algorithms. Using those algorithms, the AI ML system can learn the local statistical of legitimate data and users. Inputs that fall outside this learned distribution can be flagged as suspicious and excluded from training or decision-making.

6.4.4 Multi-agent validation

Multi-agent validation is used in distributed AI ML decision making systems. Here, rather than relying on a single learning agent, the system leverages multiple independent agents, each with its own observations and models to cross-verify spectrum decisions.

6.5 Privacy-preserving ML

6.5.1 Differential Privacy (DP)

Differential privacy is a mathematical framework designed to protect individual data contributions in machine learning systems, even when adversaries have access to model outputs or aggregated statistics. In spectrum sharing, it ensures that sensitive information like a secondary user's location, transmission history, or a primary user's activity pattern cannot be inferred from the behaviour of an AI/ML model.

6.5.2 Secure Multiparty Computation (SMC) and homomorphic encryption

SMC allows multiple parties (e.g. secondary users, spectrum access systems) to jointly compute a function over their private inputs without revealing those inputs to each other. For example, several SUs can collaboratively train a spectrum-sensing model or compute interference metrics without exposing their individual sensing logs or location data.

6.6 Summary

In conclusion, while AI/ML enhances spectrum sharing with dynamic adaptability, it also widens the attack surface across the ML workflow, protocol layers, and privacy boundaries. A multi-pronged defence which combines adversarial hardened models, cryptographic privacy guarantees, secure protocol design, and robust cross-layer auditing, is essential to ensure safe, efficient, and private spectrum sharing.

7 Conclusion

The analysis conducted throughout the present document highlights the following trend: AI/ML has the capability to improve the performance of future coexistence and spectrum sharing frameworks. From the conducted survey emerges a growing recognition that conventional, static or rule-based sharing approaches struggle to accommodate the accelerating complexity spectrum sharing. As a consequence, AI/ML could improve the sharing capabilities while keeping incumbents interference free.

From a technical perspective, the present document suggests that AI/ML can substantially improve several building blocks of spectrum sharing, in particular related to interference prediction and management. Advances in learning-based spectrum sensing, adaptive prediction models, and reinforcement learning policies for dynamic allocation indicate that AI systems can offer improved situational awareness and more responsive decision-making than classical approaches. As a consequence, this approach opens possibilities for coexistence regimes that are both more efficient and more responsive to changing real-time conditions. The aforementioned potential, however, is accompanied by non-negligible challenges. The architectural changes introduced by AI/ML elements into spectrum management gives rise to new risks within the sharing landscape. Specific vulnerabilities, from data poisoning to model extraction, introduce possibilities of attacks that are not present in traditional systems. Moreover, the increasing reliance on data can introduce concerns where sensing data, logs, or model could reveal sensitive information about both primary and secondary users. While these risks do not negate the benefits of AI/ML, they underline the need for robust security and privacy processes.

In conclusion, the findings of the present document suggest that AI/ML-assisted spectrum sharing is increasingly relevant and can improve the flexibility and performance of spectrum sharing systems. However, the high cost of AI/ML models coupled with the increased privacy risks that appropriate safeguards demand, and the reduced interest of some major administration in spectrum sharing suggest that it might be premature to investigate such systems in the short term.

History

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